Class 31: Brillouin Zones, Diffraction, and Allowed Energy Levels

In the last class we defined Bragg planes, Wigner-Seitz cells and Brillouin Zones. We also looked at specific examples of Wigner-Seitz cells and Brillouin zones, largely in two dimensions.

In this class, we will begin by looking at Wigner-Seitz cells and Brillouin zones in three dimensions, and later begin to look at the interaction of Brillouin zones and wave vectors of nearly free electrons.

As indicated in the previous classes, a simple cubic structure in real space, transforms to a simple cubic structure in reciprocal space. Further, the Wigner-Seitz cell for a simple cubic lattice is a simple cube as well. Therefore, the first Brillouin zone for a material that has a simple cubic structure in real space, is the Wigner-Seitz cell corresponding to its reciprocal lattice, and is therefore also a cube.

Consider a BCC lattice. It consists of a lattice point at the body center, and lattice points at the corners of the cubic lattice. Since the body center is conveniently at the center of the unit cell, we choose it as the central point around which the Wigner Seitz cell is identified. By treating the body center as the origin, imaginary lines can be drawn to the corner lattice points, which are the nearest neighbors of the lattice point at the body center. Perpendicular bisectors to these imaginary lines can be drawn, and the region in space bounded by these perpendicular bisectors will then form the Wigner-Seitz cell of the BCC lattice. The Wigner-Seitz cell of a BCC lattice is shown in the Figure 31.1 below.

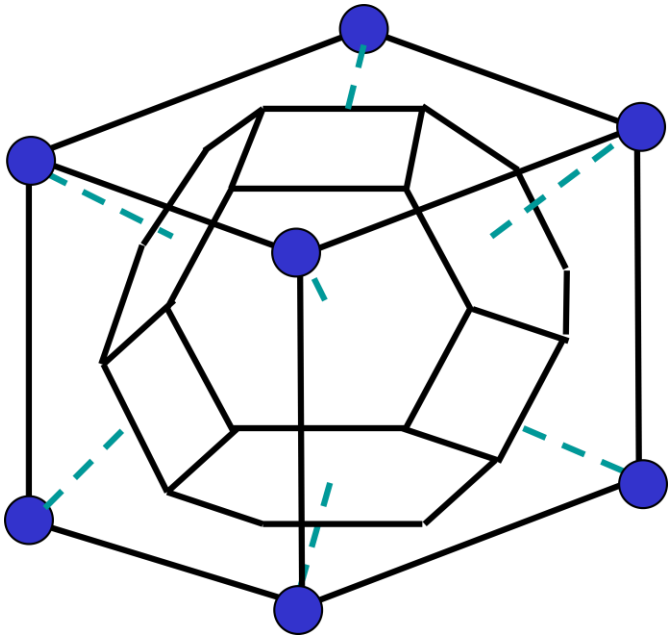
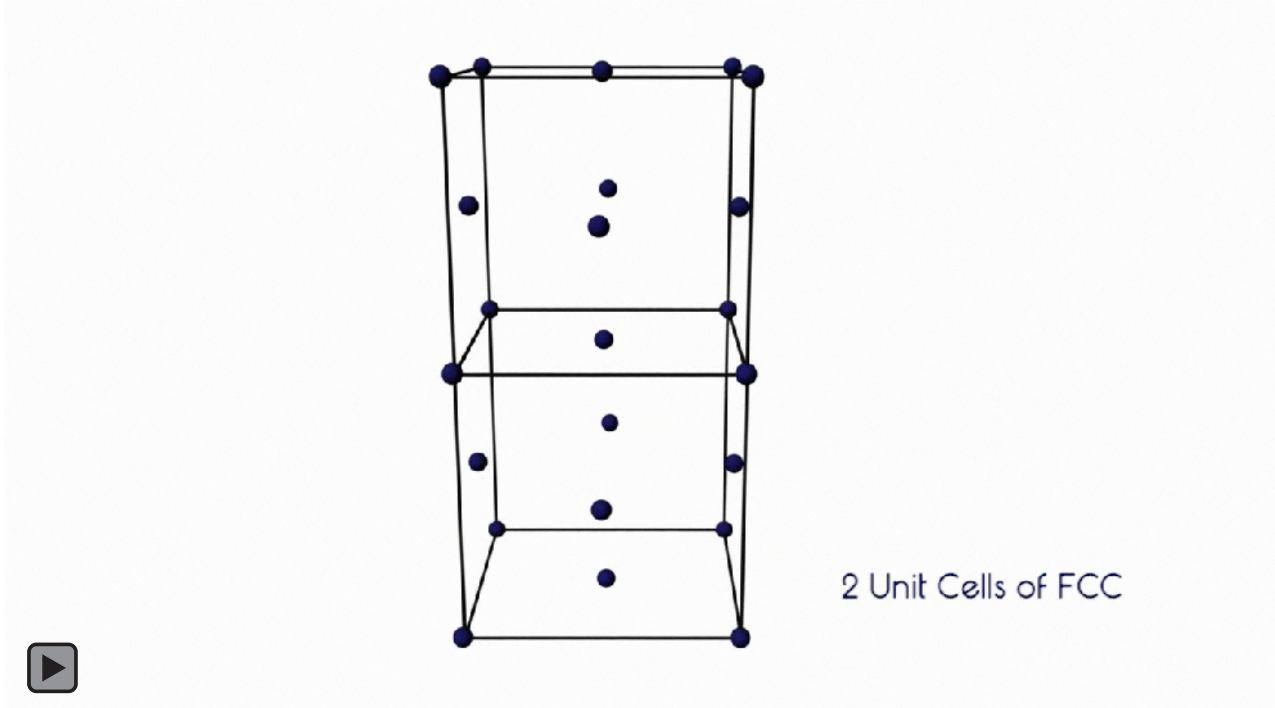


Figure 31.1: Wigner-Seitz cell of a BCC lattice. The central lattice point of the BCC lattice is hidden within the Wigner Seitz cell.

We can also consider imaginary lines connecting the body center of the unit cell chosen to its adjacent body centers. The perpendicular bisector to these lines will be the sides of the unit cell, since they will be midway between the adjacent body centers. These planes will not cut out any portion from the Wigner-Seitz cell we have identified above. Therefore the Figure 31.1, correctly identifies the Wigner-Seitz cell for a BCC lattice.

Consider now an FCC lattice. Since lattice points are present at the corners of the cube as well as at the face centers, at first glance it is not possible to identify a convenient central point about which the Wigner-Seitz cell can be defined. However by taking two adjacent unit cells together it is possible to identify the shared face center as a common central point about which the Wigner-Seitz cell can be defined. In view of the symmetry of these structures, the choice of origin is only a matter of convenience. The choice of origin, is as shown in the Figure 31.2 below.



Animation of 3D Wigner-Seitz Cell

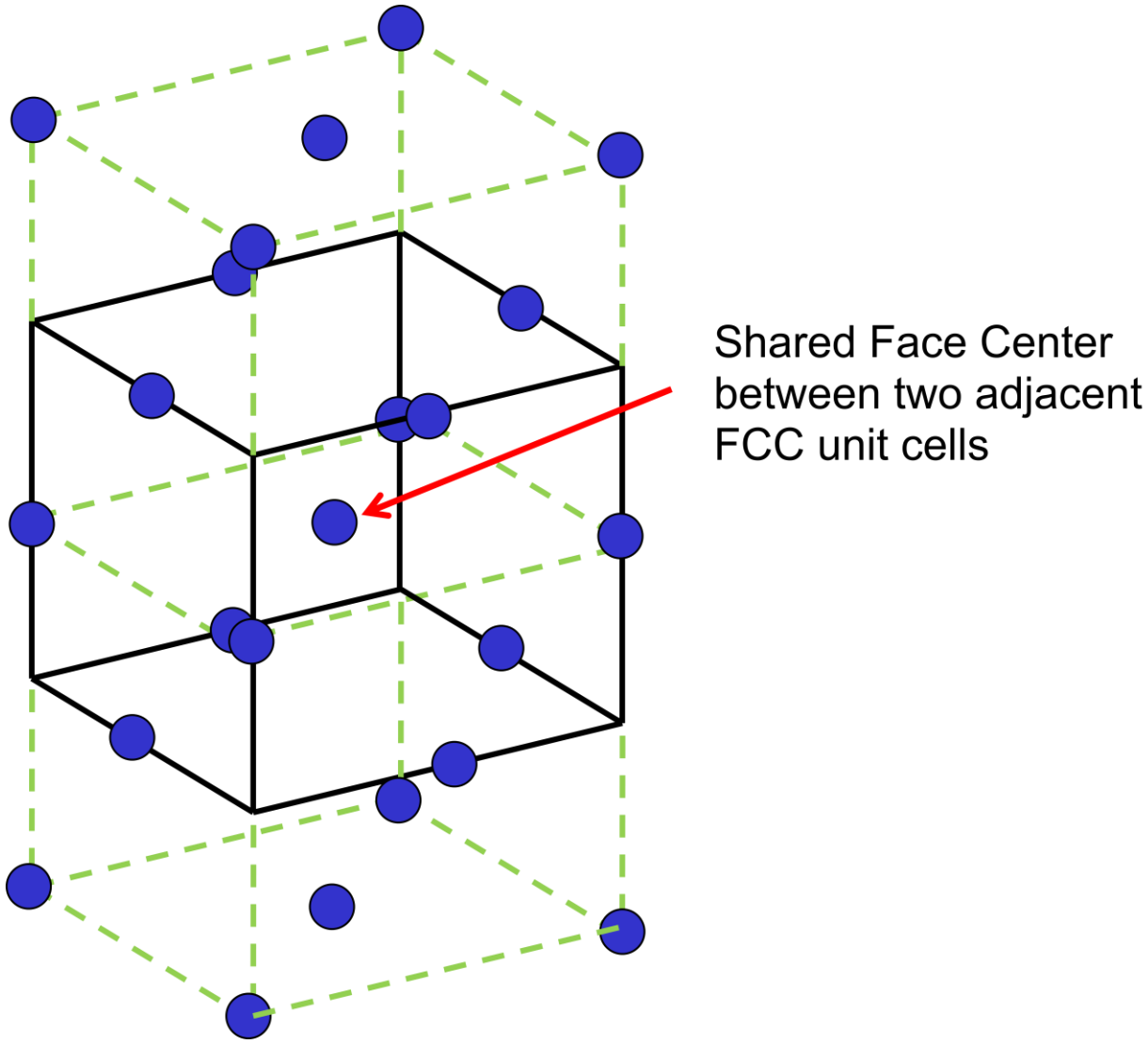


Figure 31.2: Choice of origin for an FCC unit cell, for the purpose of identifying its Wigner-Seitz cell. A face center, shared between two adjacent FCC unit cells, is the origin of choice.

The nearest neighbors of the face centered lattice point, are four face centers of the unit cell on top, four face centers of the unit cell below, and the four shared corner lattice points of the two unit cells. By drawing imaginary lines to these nearest neighbors and drawing perpendicular bisectors to those lines, the Wigner-Seitz cell of the FCC lattice is identified, as shown in Figure 31.3 below.

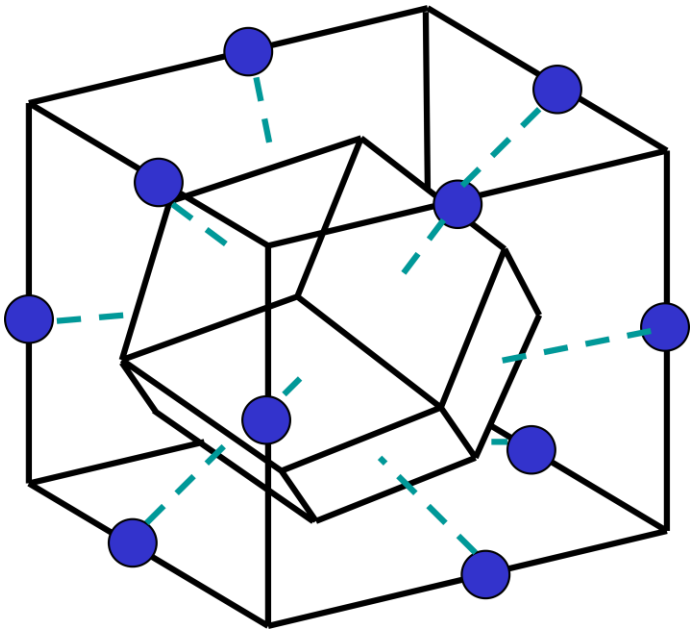


Figure 31.3: Wigner-Seitz cell of an FCC lattice. One shared face center lattice point is hidden within the Wigner Seitz cell.

Since a BCC real lattice is represented in reciprocal space by an FCC lattice, as seen in class 29, the Wigner-Seitz cell of the FCC lattice becomes the first Brillouin zone of the BCC real lattice. Similarly, since an FCC real lattice is represented in reciprocal space by a BCC lattice the Wigner-Seitz cell of the BCC lattice become the first Brillouin zone of the FCC real lattice. These structures are therefore conceptually inversely related as indicated in Figure 31.4 below.

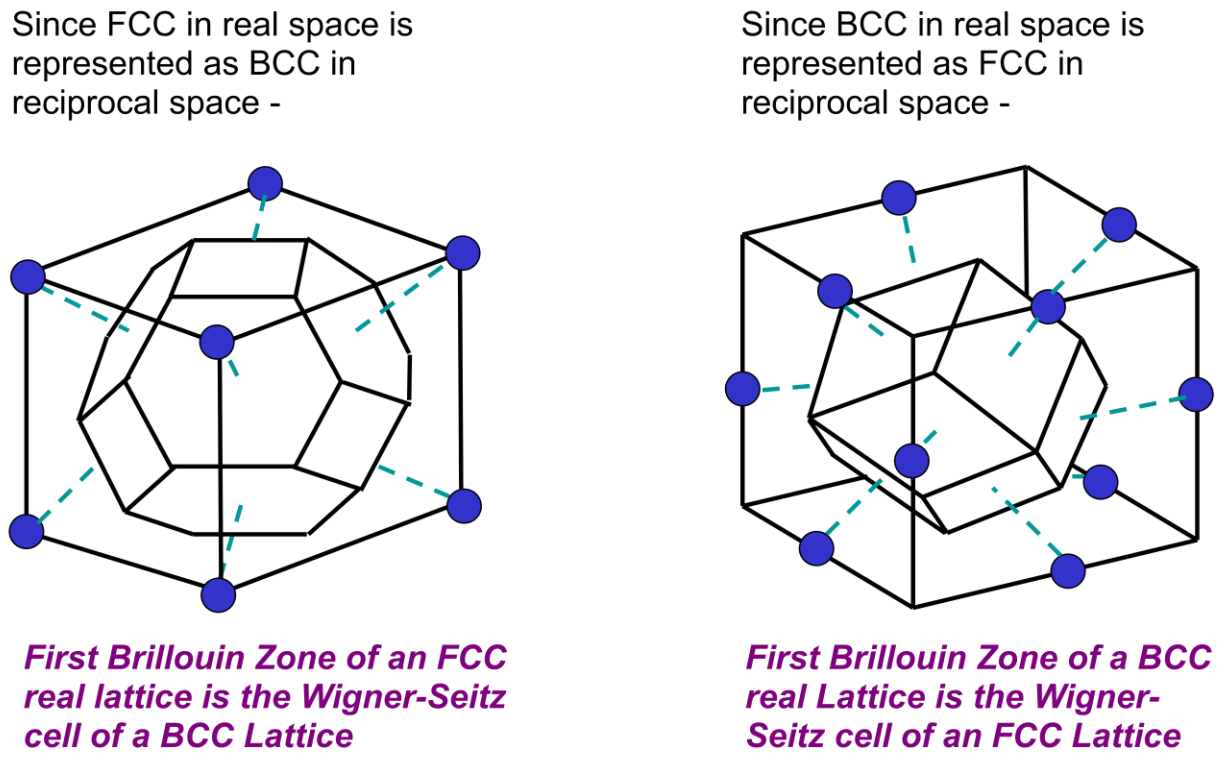


Figure 31.4: The relationship between real lattice structures and their Brillouin zones for FCC and BCC real lattice structures.

We have also noted that in the manner that the Brillouin zones are identified, their boundaries are Bragg planes. Therefore, if it turns out that specific phenomena occur at Bragg planes, then these phenomena can be expected to occur at the Brillouin zone boundaries as well.

Let us re-examine the description we have seen for diffraction in reciprocal space, and identify the significance of Bragg planes in that description.

In reciprocal space the condition for diffraction is written as

$$\frac{\vec{S}}{\lambda} - \frac{\overrightarrow{S_0}}{\lambda} = \overrightarrow{H_{hkl}}$$

A schematic of this relationship is shown in Figure 31.5 below.

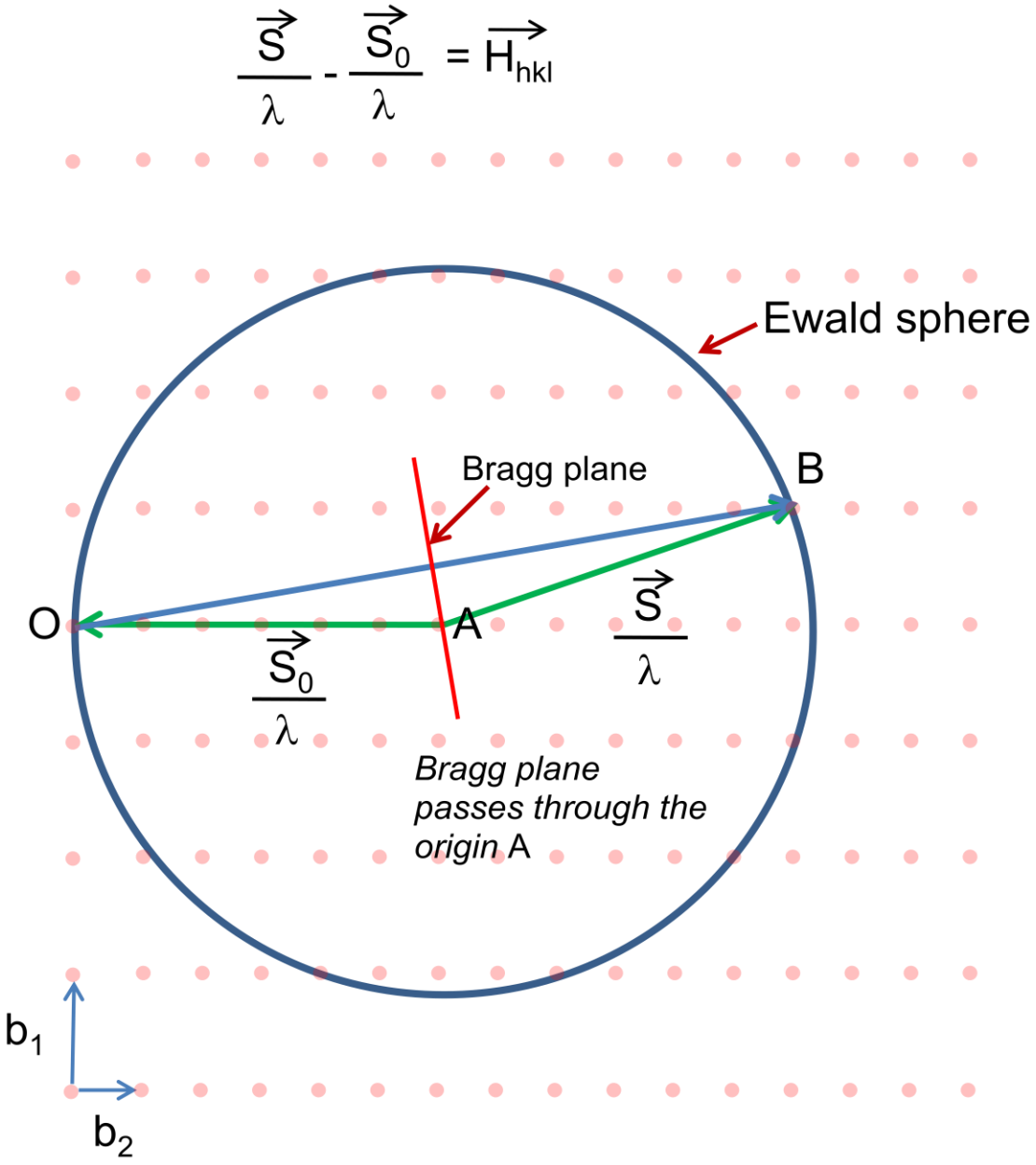


Figure 31.5: Schematic of the diffraction relationship in reciprocal space showing that the corresponding Bragg plane passes through the origin and therefore touches one end of the wave vector

If we look at the plot of this relationship, we note that purely due to geometric considerations, the perpendicular bisector of $\overrightarrow{H_{hkl}}$ will pass through the origin. This occurs because $\overrightarrow{H_{hkl}}$ is a chord to the circle drawn. In reciprocal space we have called these perpendicular bisectors of reciprocal lattice vectors $\overrightarrow{H_{hkl}}$, as Bragg planes. Therefore Bragg planes of those reciprocal lattice vectors which lie on the surface of the Ewald sphere, and hence satisfy the condition for diffraction, pass through the origin of the Ewald sphere, where they touch one end of the wave

vector $\frac{\overrightarrow{S_0}}{\lambda}$. Or stated in another manner, whenever a Bragg plane touches one end of a wave vector, automatically the rest of the diagram 'falls in place' and the condition for diffraction is satisfied, with respect to the Ewald sphere corresponding to that specific wave vector.

As mentioned earlier, the process of diffraction is not impacted by the source of the waves. As long as the condition for diffraction, as indicated by the diffraction equation, is satisfied, diffraction will occur.

Nearly free electrons in a material are roaming through the material displaying wavelike behavior and we have studied how to associate wave vectors with these electrons based on the momenta they possess. Since they roam through a material that has a periodic structure, like most metals do, there are Brillouin zones associated with the periodicity of these materials. Therefore, if we draw a consolidated plot that shows the wave vectors allowed for the free electrons, and the Brillouin zone boundaries corresponding to the structure of the material, then in this consolidated plot, whenever the wave vectors touch the Brillouin zone boundaries, diffraction will occur. The only additional condition being that the wave vectors as well as the Brillouin zones should be drawn to the same scale on that plot.

So far we have spoken about real space and reciprocal space. With a scaling factor of 2π, reciprocal space can be converted to 'k space' as follows:

Consider a one dimensional lattice of spacing 'a', the corresponding reciprocal lattice has a spacing $\frac{1}{a}$. Since k space plots λ as $\frac{2\pi}{\lambda}$, for the scaling to be the same, real lattice spacing of 'a' should be plotted as $\frac{2\pi}{a}$ in k space, which is dimensionally the same as reciprocal space.

It is important to draw our attention to one detail at this stage since it will directly impact the details in the consolidated plot we are going to draw. The lattice has a spacing 'a', which is of the order of 10^{-10} m. Therefore the corresponding k space spacing of $\frac{2\pi}{a}$ is relatively large. On the other hand the extent of confinement of the nearly free electrons 'L' is of the order of meters, or the extent of the solid. Therefore the corresponding allowed wave vectors, $\frac{n\pi}{L}$, are very closely spaced. In fact we can expect of the order of 10^{10} allowed wave vectors in the scale defined by two adjacent lattice points in k space. The allowed wave vectors, though discrete, will look continuous on this plot simply as a result of the scale of the plot.

Since the reciprocal lattice vectors are $\frac{2\pi}{a}$ in length, the Bragg planes for this reciprocal lattice occur at $\frac{1}{2}\times\frac{2\pi}{a}=\frac{\pi}{a}$.

Therefore, starting at the origin, Brillouin zone boundaries occur with a spacing of $\frac{\pi}{a}$.

The extent of the Brillouin zones for this lattice are therefore as follows:

First Brillouin zone: $-\frac{\pi}{a}$ to $+\frac{\pi}{a}$

Second Brillouin zone: $-\frac{2\pi}{a}$ to $-\frac{\pi}{a}$ and $+\frac{\pi}{a}$ to $+\frac{2\pi}{a}$

n^{th} Brillouin Zone: $-\frac{n\pi}{a}$ to $-\frac{(n-1)\pi}{a}$ and $+\frac{(n-1)\pi}{a}$ to $+\frac{n\pi}{a}$

The Brillouin zones of this one dimensional lattice are shown in Figure 31.6 below:

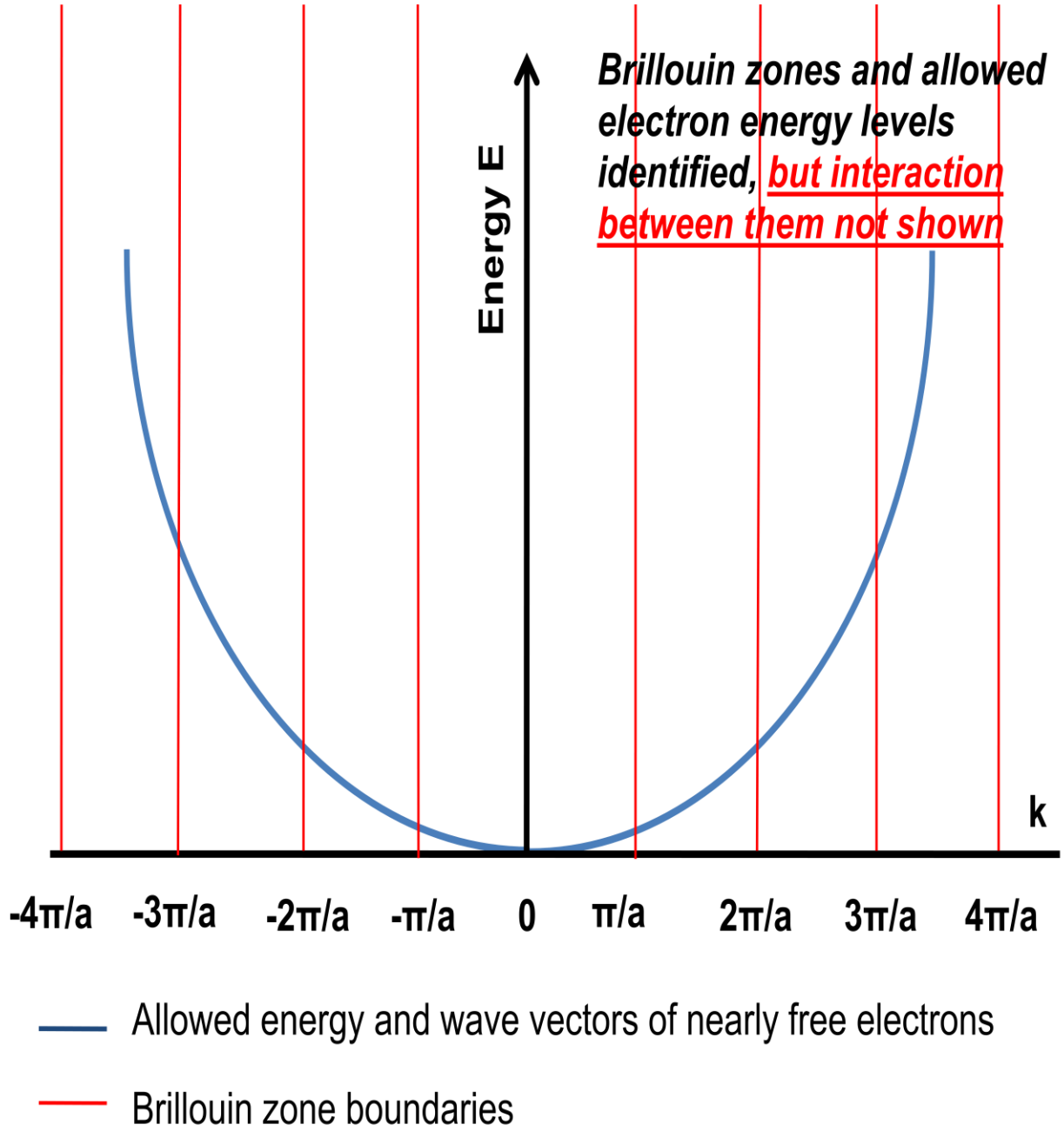


Figure 31.6: Brillouin zones corresponding to a one dimensional real lattice of spacing '*a*', and wave vectors corresponding to nearly free electrons, drawn to the same scale in *k* space. The interaction between the Brillouin zones and the electron wave vectors ***has not been shown*** in this figure. The figure primarily highlights the relative scales of the two pieces of information that have been plotted. The discrete allowed wave vectors of the electrons look continuous when drawn to this scale.

In the next class we will examine the interaction of the allowed wave vectors and the Brillouin zones and realize that this interaction results in forbidden values of energy for the electrons, and hence the origin of the band structure.